Real-time Fall Detection System Based on Deep Learning

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ABSTRACT

- **Taking care of health for the elders is an essential part of the social policies in numerous countries.**
- **The supervising falls in the elderly is challenging work in a long time ago while it takes time, money, and concentration all the time that also deprives from private space.**

In this research, we have applied YOLOv4-tiny approach with Raspberry (an embedded computer) system to create a detected falling system. A camera is utilized to detect objects and do calls and send messages to the family of falling.

Keywords— Falls in elderly , Fall Detection, YOLO and Raspberry.

INTRODUCTION

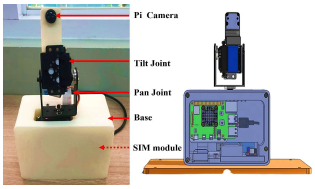
What are the problems associated between the elderly and falls?

- According to statistics from WHO in 2021 [1] that falling are the second leading cause of unintentional injury deaths worldwide.
- Each year, an estimated 684000 individuals die from falls globally of which over 80% are in low- and middle-income countries.
- Adults older than 60 take the largest number of fatal falls compared with the other ages.
- 37.3 million falls that are severe enough to require medical attention.
- An estimated 684000 fatal falls occur each year.
- Healing abilities in the elderly is slow, and all phases of wound healing are influenced.



Is our method different from the others?

YOLO combines Raspberry used in some other projects before. However, this system still has untapped potential, we want to update the system with servos and module SIM to increase its effectiveness.



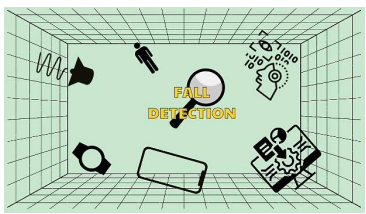
Main contribution:

- In the project, we propose a method for fall detection based on YOLOv4-tiny and embedded in Raspberry Pi 4.
- We have collected a dataset (about 4500 images) for training which is used in this paper.
- By linking the external module (SIM and camera) with the powerful embedded system (Raspberry Pi4), we hope to provide an affordable and reliable system that can spread the whole society.

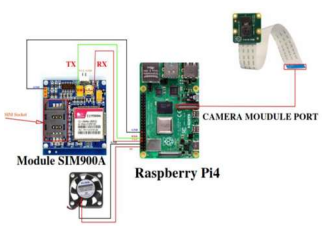
RELATED WORK

Are there any methods that are used to prevent or detect falls ?

- Using multimodal data [2].
- 4-D mmWave Radar and a Hybrid Variational RNN AutoEncoder [3].
- Accelerometer in smart gadgets
- Using CNN algorithms [4].

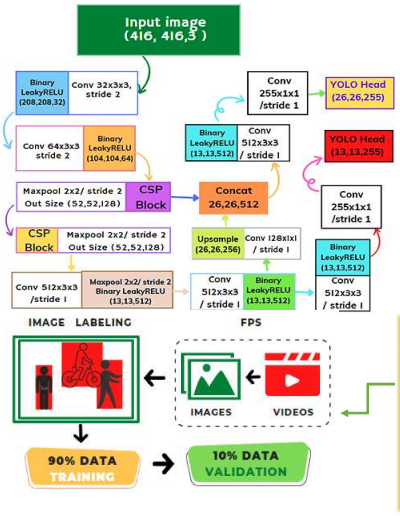


PROPOSED METHOD

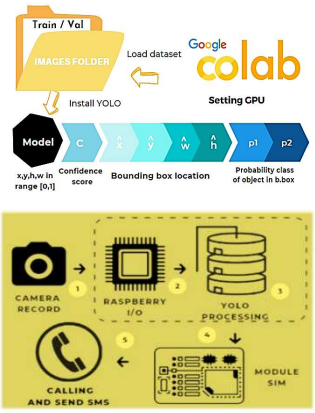


What we have done in the project?

- Design and implement a falling detection system.
- Transfer data between the SIM module and embedded system.
- Detect falling and non-falling person
- Data collecting.

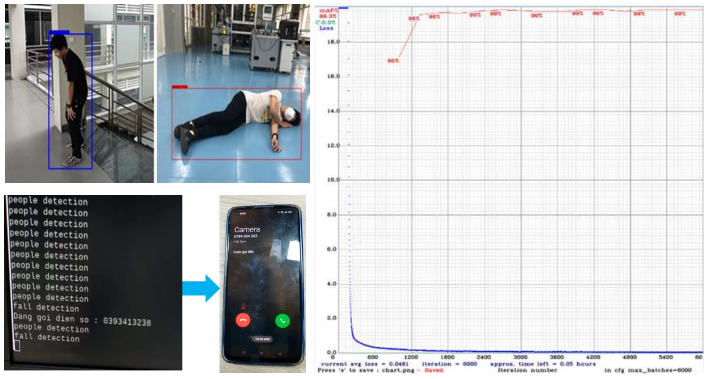


- Use YOLO V4 Tiny architecture to train fall detection model
- Training model process



EXPERIMENTAL RESULT

- Evaluation of our training data with YOLOv4-tiny.
- Results of our fall detection based on YOLOv4-tiny, the left image is a people detection in a blue box, and the right image is a fall detection in a red box.
- The system will do a call to mobile when a falling case is detected.



CONCLUSION

- Increasing the amount of elderly means increasing the fatal fall ratio and how important fall detection is.
- A good detection system can seriously reduce and prevent us and our families from injuring over time.
- With the custom dataset and training model which was experienced, the proposed method has a good performance and accuracy rate of falling detection. However, our model is in the early stages of following people who fall so that there are many things will be needed to do to get the best system.
- In the future, we hope that our system can follow people, and detect human behaviors.

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